E3.1.1 Energy-conversion efficiency

Efficiency is useful output energy divided by total input energy, or corresponding powers over the same interval. Ordinary conversion efficiency does not exceed 100%. The remaining energy goes into forms outside the intended useful output. Define what counts as useful and convert percentage efficiency to a fraction before substitution.

$$\eta_{\%} = \frac{E_{\text{useful}}}{E_{\text{input}}} \times 100\%$$

Percentage efficiency; useful output energy and total input energy in identical units. A fractional efficiency omits × 100%.

E3.1.2 Useful heating power

For water of mass m warmed by ΔT in time t with no state change, useful heat is $Q = mc\Delta T$ and useful power is $P_{\text{useful}} = \frac{mc\Delta T}{t}$. With efficiency η, input power is $P_{\text{input}} = \frac{mc\Delta T}{\eta t}$, since efficiency is useful output divided by input. Convert minutes to seconds and use consistent mass and specific-heat units. With losses, water heating power is lower than appliance input power.

E3.1.3 Interpreting energy labels

The textbook's energy-label example uses grades 1–5, with grade 1 most efficient and grade 5 least efficient. Energy use and operating time can also be used to calculate input power. Efficiency grade and total electricity consumption are different pieces of information. Compare within the product category and conditions provided in the question.

E3.1.4 Hotplates, induction cookers and microwave ovens

A hotplate heats a resistive element and transfers energy to a pan. An induction cooker uses a changing magnetic field to induce currents and heating within a suitable metal pan. A microwave oven's oscillating electric field drives polar molecules in food, with interactions converting energy into internal energy. Their heating locations and mechanisms differ.

Electric hotplate	Induction cooker	Microwave oven
Working principle		
heating effect of current in coil → conduction / radiation to pot → conduction to food	heating effect of eddy currents induced in the metal base of cooking pot → conduction to food	flipping of water molecules in food due to microwaves, producing heating effect → conduction within food
Advantages		
• most pots with flat bases are suitable to use • low selling price	• high energy efficiency • top plate does not get very hot	• ideal for reheating (Fig 1.2s) • its case does not get very hot
Disadvantages		
top plate gets very hot	pots made of ferromagnetic materials are usually required	low energy efficiency

Table 1.2a *Comparing different types of electric cookers.*

Textbook Table 1.2a · p.18

E3.1.5 Energy flow in a heat pump

A heat pump uses external work to move energy from a colder region to a hotter one, opposite natural heat flow. Heat delivered to the hot side equals heat removed from the cold side plus work input. A refrigerator releases more heat at its back than it removes inside. It does not create energy.

$$Q_H = Q_C + W$$

Refrigerator/heat pump per cycle: Q_{H}: heat delivered to hot reservoir; Q_{C}: heat removed from cold reservoir; W: work input. All are positive energy magnitudes (J).

The four components of the refrigeration cycle

Refrigerant passes through the evaporator coil, compressor, condensing coil and expansion valve in a repeating cycle that transfers heat from indoors to outdoors. The table follows the textbook's simplified model and shows the state changes and processes in each component.

Component	State change (inlet → outlet)	Process and function
① Evaporator coil	Liquid + gas → gas	Cold refrigerant absorbs latent heat of vaporization from indoor air. The liquid portion evaporates into gas, cooling the room. The refrigerant warms but remains colder than the room.
② Compressor	Gas → gas (no change of state)	The compressor does work on the gaseous refrigerant, increasing its pressure and temperature until it is hotter than the outdoor air.
③ Condensing coil	Gas → liquid	Hot gas releases latent heat of vaporization to the outdoor air and condenses into liquid. The refrigerant cools but remains hotter than the outdoor air.
④ Expansion valve	Liquid → liquid + gas (partial evaporation)	Pressure drops after the valve and some liquid evaporates. The energy for evaporation comes mainly from the refrigerant itself, cooling it below room temperature before it returns to the evaporator coil.

Cooling capacity and coefficient of performance

Cooling capacity is heat removed from the cold side per second. Refrigerating coefficient of performance is heat removed divided by work input, or the corresponding power ratio. It can exceed 1 because the input work moves existing heat. Compare heat and work over the same time interval.

$$P_{\text{cool}} = \frac{Q_C}{t}$$

$$\text{COP}_{\text{cool}} = \frac{P_{\text{cool}}}{P_{\text{in}}} = \frac{Q_C}{W}$$

Cooling capacity P_{cool} (W); Q_{C}: heat removed from the cold reservoir (J); t: s. Not the electrical input power.

Cooling coefficient of performance is dimensionless and may exceed 1. Q_{C}: heat removed; W: work input, both in J and over the same interval.

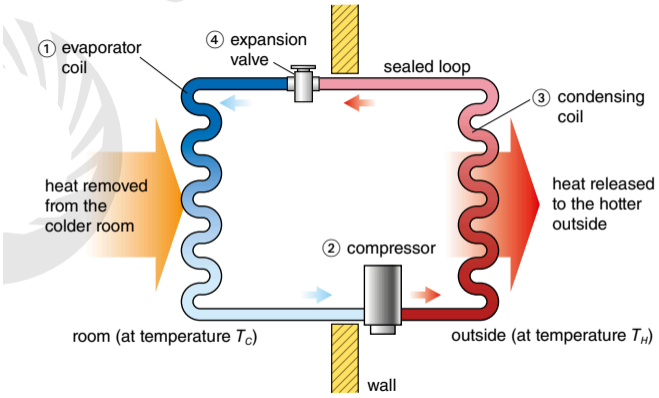


Fig 1.2d Cyclic process in an air conditioner

The evaporator absorbs heat, the compressor receives work, the condenser releases heat and the expansion valve lowers pressure.